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ANALYSIS OF MATCH QUALITY IN LEAGUE OF LEGENDS ACROSS DIFFERENT TIMES OF THE DAY

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Abstract

This bachelor thesis investigates the influence of time of day on match quality in League of Legends, a popular online game. It explores how players' performance in League of Legends varies across different time periods, considering factors like circadian rhythms and daily activities. By leveraging Riot Games' API, the study analyzes a large dataset and implements LASSO regression, to predict match quality. The results show significant variations in gameplay quality, with early afternoon being optimal for the most competitive games, and late-night to early morning sessions showing decreased quality. This study provides insights for players looking for optimal gaming times and contributes to understanding online gaming behavior in relation to time of day.

Keywords

League of Legends, LASSO Regression, match quality analysis, fundamental gameplay variables, chronobiology in gaming, time of day effects, performance variability

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Introduction

League of Legends, a flagship title by Riot Games, is the leading video game in the genre of multiplayer online battle arenas (MOBAs), which receives increasing interest year after year. I aim to explore a nuanced, often overlooked aspect of this game - the impact the hour of the day has on match quality.

The primary purpose of my research is to determine whether the time at which players choose to participate in League of Legends matches affects their in-game performance. This hypothesis is based on the fact that our cognitive and physical capabilities fluctuate throughout the day. These changes are influenced by our circadian rhythms, the daily tasks we undertake, and the personal lifestyle choices we make. I suspect that these factors may have a noticeable impact on players' performance. League of Legends is ideal for this type of analysis, as it can be played online with virtually no limitations.

The first phase of this study defines what makes a "high-quality" League of Legends match. Subsequently, I plan to analyze game data across various time frames to identify patterns or discrepancies between individual hours. I hope to provide fresh insight into the relationship between players' daily routines and their game performance variability.

The main goal of my thesis is to analyze the quality of individual matches using post-game statistics in a regression model. I hypothesize that as the level of play increases, certain variables will reveal patterns that reflect the average game quality. The model will predict the quality of the game based on these variables. I aim to explain the differences between the predicted and actual game qualities by examining the influence of the time of day and identifying peak hours when match quality typically increases or decreases.

Furthermore, my research will focus on identifying and examining the key variables that influence match quality in League of Legends. Understanding these critical aspects of the game can be intriguing, particularly for players aiming to improve and climb the ranking ladder. To achieve this, I will simplify the original regression model to a more basic form with fewer predictors. This step will help me focus on the key variables that significantly influence the quality of a match.

By combining perspectives from chronobiology and online gaming, this thesis aims to provide a new outlook on game studies. I will attempt to reveal how the time of day can shape the gameplay experience in League of Legends. The findings could in practice serve as a framework for players to select their preferred time slot for their gaming sessions.

1 Effect on human performance based on time of the day

Since forever, people have had preferences for performing certain endeavors at particular times of the day that even led to changes in their bodily functions (O'Keefe, 2023). It could be as basic as sleeping during the night (Cohen et al., 2010) or, on the other hand, more nuanced activities, like peak creative thinking (Wieth & Zacks, 2011), which may be more personalized and not apply to all individuals. Multiple studies have been conducted trying to associate time of the day with performance (Lenné, Triggs & Redman, 1997; Hines, 2004; Laird, 1925). But even without these studies, we can see those signs in ourselves.

Humanity has exploited people's circadian rhythm since the dawn of time, a practice that is especially evident in today's shift work (James et al., 2017). Related to this phenomenon, in times of war, it is extremely common to start an attack during the night with the intent of surprising the enemy and having an upper hand right at the beginning (Sasso, 1983). While League of Legends is a voluntary activity that people choose to engage in, often as a hobby, it can also lead to addiction for certain individuals, as researched by Weinstein (2010). Resulting in continuous playing late into the night, thus sleep deprivation having an impact on the quality of their performance in the game - especially their reaction time (van den Berg & Neely, 2006).

Many people consider themselves „early birds“, meaning they prefer to wake up early in the morning, feel energized after waking up, complete important tasks early in the day, relax during the evening, and go to bed relatively early. On the flip side, there are individuals who perceive themselves as "night owls". They are basically the opposite of early birds. They wake up and stay up late, schedule their activities later in the day, and are a bit languid during the morning while gaining focus later in the afternoon. Several studies have been carried out in an effort to compare either hormonal levels in these two types or performance in specific tasks (Lepone & Yang, 2020; Gariépy et al., 2019).

While my study does not necessarily focus on the differences between early birds and night owls, they certainly play at least a partial role in the analysis. It would be very out of the ordinary for a morning person to play a League of Legends game at 12 AM and for an evening person at 8 AM.

A part that is a bit problematic for my subsequent time analysis is that the popularity of playing League of Legends is not equally distributed throughout the day. This stems from people preferring to play video games after they have already completed their errands and other mandatory tasks, thus leaving League of Legends as a relaxing activity to wind down at the end of the day (Cummings & Vandewater, 2007).

There is also a very interesting phenomenon showing that people's performance in highly mentally stimulating activities decreases after an extended period of time (Boksem, Meijman & Lorist, 2005). As stated before, people tend to play League of Legends later in

the day, meaning games played after midnight could very well be from players that have already been playing since the evening, thus lowering their performance in the game.

It was also reported that people's mental capacities are affected by their food consumption (Mahoney, Taylor & Kanarek, 2005). Usually, after a meal, their cognitive performance drops, resulting in a so-called post-lunch dip (Monk, 2005).

Studies have shown that while the time slot during which the peak for cognitive abilities arises varies based on individuals, for adolescents and young adults – the overwhelming majority of League of Legends players (Tyack, Wyeth,& Johnson, 2016) – they reach their peak during the afternoon or evening (Facer-Childs, Boiling & Balanos, 2018). As mentioned before, League of Legends is certainly a complex mental exercise. It will be interesting to see if I happen to confirm or deny this phenomenon.

2 Brief introduction to League of Legends

League of Legends was published by Riot Games in the year 2009 as its first and, until 2019, its only title. League of Legends is one of the most popular online video games in the world and certainly the leading MOBA (multiplayer online battle arena), a genre in which its only real competitor is DotA 2.

Unfortunately, video game publishers (including Riot Games) are not keen on sharing data about their users, meaning I am forced to rely on third-party data sources, which do not share their methodology on their website. It can be estimate that League of Legends has approximately 153 million individual accounts every month (Priori Data, 2023). This number does not necessarily reflect the amount of unique users as it is quite common for players to own more than one account or, as an alternative, albeit less frequent and certainly against Riot's Terms of Service (Riot Games, 2023), have a shared account with other players.

One aspect in which League of Legends is certainly thriving is its esports popularity and viewership. People enjoy watching high-stakes League of Legends matches even though they might not play much themselves nowadays. Five out of the six most-viewed esports matches are from League of Legends events, with the highest peak at 6.4M (Esport Charts, 2023).

League of Legends is actively updated every two weeks with minor changes and twice a year with more impactful, game-changing patches. This keeps the game fresh and appealing to its player base.

2.1 Gameplay introduction

For my following analysis, it is recommended, albeit not necessary, to understand the basic mechanisms of League of Legends. In this chapter, I try to summarize the most fundamental parts of the game, so readers are, for the most part, on the same page. Of course, more advanced knowledge of League of Legends can be beneficial for understanding and mainly deriving practical applications from my study.

By a large margin, League of Legends' most popular and only competitive map is Summoner's Rift. In which there are two teams each consisting of five players with the ultimate goal being to destroy the opponent's team Nexus – its destruction results in a victory. Before getting toward the Nexus, players must destroy other structures that protect it – towers and at least one inhibitor.

Subsequent objective in League of Legends involves trying to accumulate gold and experience, which makes champions stronger in fights against the opposing team. That comes mostly from killing enemy minions, which spawn in waves every 30 seconds in each of the 3 lanes at Summoner's Rift, and from jungle camps, which spawn in the jungle on

both sides of the map, as well as from enemy champions. Killing an enemy champion results in the temporary disappearance of that champion from the map and rewarding the other team with the opportunity to push or take objectives more freely. Players are able to spend their gold in the shop to purchase items that make their champion stronger. Accumulating experience leads to gradually increasing the level of the player's character up to the cap level 18. Every level grants base stats making them stronger but also ability points, allowing them to upgrade one of their four abilities. Abilities always scale with their level, making them stronger in some way, for example dealing more damage or reducing their cooldown.

Another way to strengthen a team's position in the game is through capturing neutral objectives. There are 3 types of these in League of Legends. Rift Herald spawns in the river between the top lane and middle lane up to two times before the 20-minute mark. Upon capturing Rift Herald, the slayer gets an influx of gold and the opportunity to pick up the Eye of Rift Herald, which, if summoned, can charge towards a turret and deal a significant amount of damage to it. After 20 minutes, Baron Nashor spawns at the place of the Rift Herald instead. The team that slays Baron Nashor gains a decent amount of gold and experience, but most importantly, a temporary baron buff, that allows them to push with more ease as their minions are stronger during that timeframe. Baron Nashor respawns every 6 minutes after its death. The last neutral objective is Dragon, which spawns between the middle lane and bottom lane every 5 minutes upon its capture. The team that kills the dragon gains a permanent buff of a various kind depending on the type of the dragon. A team that accumulates 4 dragons gains a soul that is very powerful and lasts until the end of the game. After a soul has been reached, a special type of dragon spawns, the so-called Elder Drake. The team that is able to slay the elder drake gains a temporary automatic execute against enemy champions if they are below 20% of their maximum health. This buff is considered the strongest in the game.

League of Legends is designed such that a team generally cannot afford to give up all objectives and expect to win. The losing team needs to find certain times when the disadvantage is not as drastic and take their stance against the opposition.

Players, prior to loading into the game, choose their champion from among 166 characters in the champion select. These champions have different abilities players can cast and dictate the optimal way to play the game. In modern League of Legends, players on each team perform a particular position. In the early part of the game, the top laner, mid laner, and bot laner each kill minions in their dedicated lane, while the jungler kills jungle camps in the jungle. The support is usually with the bot laner, also helping other players around the map from time to time. Later in the game, players tend to group more together when only the top laner is split from the team. Other members form more of a unit and try to progress the game towards victory. Often times, neutral objectives or pushes that threaten an end to the game are fought in 5v5 settings called teamfights. The outcome of teamfights is what generally decides the winner of the game because victorious teamfights give the opportunity to take different objectives on the map.

What is magical about League of Legends is that every champion is specific and unique. That leads to substantially different playing experiences every game, regardless of the particular combination of characters.

2.2 Ranking system

League of Legends uses a ranking system to distribute players depending on their skill level. In total, there are 10 tiers. The first 7 tiers are split into 4 divisions each, while the last 3 tiers, the so-called apex tiers, show the exact position of the player on the server. Players use the ladder as a motivation to improve in the game and, in turn, to climb higher, creating an infinite loop as most players never become satisfied with their current rank.

Players can progress within the system only by winning or losing games; that is the only relevant data the system takes into account when assessing players' skill. Players are rewarded points for a win, and reaching 100 league points promotes them into a higher division. Conversely, a lost game leads to point deduction, and upon reaching 0 league points, a player demotes into a lower division. This may sometimes be frustrating as League of Legends is a team game, so the outcome of the game is not solely in an individual's hands.

Table 1 – Player distribution across ranks (League of Graphs, 2023)

Rank	% of players	Cumulatively
Challenger	0.02%	0.02%
Grandmaster	0.04%	0.06%
Master	0.51%	0.57%
Diamond	2.93%	3.50%
Emerald	13.30%	16.80%
Platinum	16.50%	33.30%
Gold	20.10%	53.40%
Silver	19.00%	72.40%
Bronze	20.00%	92.40%
Iron	7.60%	100.00%

As Table 1 indicates, only less than 1% of players are in the apex tiers (Master, Grandmaster, Challenger), with a majority of players being placed low on the ladder, creating a rather steep ladder. While Riot Games is flattening the ranking distribution curve each year, League of Legends still has the lowest amount of players inside the top ranks compared to other popular multiplayer video games.

In addition to the ranking system, League of Legends also uses skill-based matchmaking, the system aims to match players based on their MMR (matchmaking rating), an invisible metric accessible only to Riot Games. This approach of using hidden metrics for matchmaking is prevalent in many other video games (Sarkar, Williams, Deterding, & Cooper, 2017). For the sake of simplicity, I will assume that a player's rank equals their MMR. This presumption is based on the basis that over time, a player's rank aligns with their true skill level as represented by their MMR. Of course, the system is designed to find a game for players relatively quickly (so they do not lose interest). At night when the number of players is low, especially at the highest brackets, it is rare that all the players in the game are even in the same tier.

2.3 Skillset necessary for League of Legends

Players need to be well-versed in game knowledge, which includes understanding the capabilities of other champions in the game, as well as NPCs (non-player characters) units and structures. These categories are extremely broad and entail a lot of specific information players can use in the game. This part of the skillset is the biggest reason League of Legends is considered fairly unfriendly towards new players, as there is a sizeable barrier, they need to overcome in order to understand the game.

Another area in which the proficiency of players is being tested is their overall mechanical ability to execute the plays they envision. This alludes to their capability to click specific keyboard keys and mouse buttons either in reaction, at a certain time, in a specific order, or any combination of the aforementioned elements. Players who have played other video games before, especially other MOBAs, are generally able to at least partially transfer this skill into League of Legends. Improving this aspect of the skillset is hard, albeit not impossible, for most of the player base. That stems from the fact that reaction speed is heavily genetic (Luciano et al., 2004), thus limiting the extent to which mechanical skill can be enhanced.

An additional challenge posed to the players is also their ability to dynamically decide what is the correct decision in every situation presented to them. I want to emphasize that these choices are happening in real-time, meaning participants have to decide immediately. The game tests them to a degree on their game knowledge but also their instinct, as they must make the selection on the spot.

Lastly, players also have to be well-versed in social dynamics, as League of Legends is a team game where players are forced to cooperate in order to win. This category is closely linked to their mental strength, as it is essential for players to maintain composure and not give up on the game prematurely, which could be the most important aspect of a player's skill set.

I would like to also add that players of the same rank do not necessarily have to be of similar strengths in the aforementioned domains, but their overall performance typically balances out. There are players who are particularly good in the mechanical element compared to the players in their rank, but their mental approach can hinder them, and vice versa.

3 Data gathering

Just as Riot shows reluctance to share statistics regarding their player base, they also seem unwilling to provide easily accessible large post-game files. The only available files suitable for analysis are created by third-party entities. Unfortunately, they are now quite outdated. League of Legends is a constantly evolving game, so while these files could provide baseline information, it is better to have up-to-date post-match statistics so the analysis corresponds to the actual live game.

Luckily, Riot Games provides access to developers or other interested individuals to their API (Application Programming Interface) at <https://developer.riotgames.com/>. Through the API, users can download data for individual matches along with various other insightful statistics. However, the API has its limitations; it is impossible to specify certain dates, and only games newer than a year can be retrieved. These restrictions do not inhibit my project in any meaningful way because I am mainly interested in generating a large modern dataset encapsulating random matches through a wide ranked range.

The main and most comfortable way to access the Riot Games API is through multiple libraries in Python such as Cassiopeia (2023) or Riot Watcher (2023). But I wanted to make all of my analysis in R, so I found a different way. There is only one usable guide available, provided by Hallach (2021), which is now outdated. I have used his code as a backbone for creating my dataset – downloading JSON files and turning them into a list of lists. As I mentioned, the code is outdated and does not reflect the current API structure.

This leads to the first practical implication of my thesis, as code retrieving information from the Riot API in R is not anywhere publicly available in a usable form. My code allows for straightforward retrieving of fresh data with the most recent matches. Alternatively, others can use my list of lists with all the information from the API to create their own dataset with matches I have generated.

After working with the Riot API, I discovered it is not particularly effective for searching and downloading random matches, as it requires multiple queries to access a certain game. The biggest problem turned out to be generating the average rank in a specific match. The rank of a player was not mentioned in the JSON file of the post-match API request. To access the rank of every player in the game, I had to perform an additional 20 API requests, which slowed down the process dramatically.

With that being said, I want to acknowledge my gratitude to Riot Games. I have stated in my application to gain permanent access to the API that in order to generate a large enough dataset I would appreciate a higher limit than they allow for personal projects. They have accommodated my inquiry, and I have received even a 10-fold higher rate request limit than the standard. This significantly reduced the time needed to obtain the dataset, for which I am really thankful.

To improve the data from the API, I made two modifications. Firstly, the rate limit is only applicable per route. There are four routes for the regions League of Legends is able to be played in. Therefore, to speed up the data download, I have used the most populated server from each route – that being NA (North America) in Americas, EUW (Europe West) in Europe, KR (Korea) in Asia, and lastly VN (Vietnam) in SEA (South East Asia). This alteration allowed for also a possibility to describe skill discrepancies between servers, as this has always been a topic of interest in the League of Legends community.

I also aimed for a more balanced distribution between individual tiers, as otherwise, the dataset would end up with only a small portion of matches played in Apex ranks. That proved to be challenging on both ends of the spectrum, as these players are often mixed with other rank brackets as a means to save time in the matchmaking process.

As mentioned above, there is no concrete information regarding the average rank of the game. First I gathered all the ranks of players that participated in the game. Then I removed the highest and the two lowest ranks of the group to account for players with abnormally high or low MMR. Afterwards, I transformed their ranks into integers starting in Iron IV at one and increasing by one for each subsequent division. I calculated the group average and converted the number back into its corresponding division, giving me all the necessary information to complete my analysis.

3.1 Sorting process

I have successfully obtained information regarding post-game statistics, but they were in a list of lists format, which was not suitable for further analysis. I examined the data in the lists and created hundreds of columns that had the potential to impact the quality of the game. These columns were derived using my understanding of the game; the four main types were total (for the whole match), per minute (for each minute of the match), delta (the difference between the winning and losing team), and percentage (share between teammates). The columns encompassed all possible statistics available after the match.

The API was not able to gather all the information for every particular match, resulting in multiple columns being incomplete. In response, I deleted columns missing more than a third of their records, and afterwards, removed every match with incomplete columns. This reduced my final dataset to 98,726 observations of 644 variables without any missing values. The server where the match was played was treated as a dummy variable, creating binary columns for NA, VN, and KR with the base server being EUW. That led to a dataset in which all the variables for determining game quality were in numerical form.

The last decision I had to make revolved around splitting the data to achieve the goals I had set out for my research. I used 40 % of the data for the dataframe that served as a basis for the analysis of individual hours, with the caveat that every hour had to have more than 300 observations to facilitate a comprehensive and robust analysis. Of the remaining 60 %, I allotted 70 % as a foundation for model creation through LASSO regression. The remainder of the data was allocated to the last partial dataset, serving as cross-validation for judging the quality of the created model.

4 Methods

In this chapter, I outline my approach to measuring game quality, a metric that extends beyond basic post-game statistics and requires a more creative methodology.

Everything stems from the assumption that a game with a higher rank average is of superior caliber compared to a game with a lower average. This premise stands on the back of the, above-mentioned, skill-based matchmaking, as its sole purpose is to distribute players by their ability to win a game of League of Legends. Thus, matching better players together, resulting in increasing game quality as the average rank rises.

As a first step, I will transform the average rank of the game into an integer, with the lowest being in Iron IV (the lowest division in the game), while the highest is in the most prestigious tier – Challenger. Then, I will develop a model capable of predicting the rank of the match by using various post-game statistics from the game, and its effectiveness will be tested using cross-validation. I will apply said model to a new dataset, creating a new column, prediction. Afterwards, I will subtract the real average rank of the game from the prediction. Lastly, I will use these differences to find out through statistical tests which hours have a consistently higher prediction than their actual rank, labeling them as high-quality hours, and vice versa. I will attribute the unexplained variability in game quality to the time of day, thus determining the hours during which match quality improves or declines.

For the prediction model, I decided to use LASSO regression, a method firstly detailed by Tibshirani (1996). Just as a brief introduction, LASSO regression, short for Least Absolute Shrinkage and Selection Operator regression, is a variant of linear regression that adds a penalty term to the least squares method. In classical linear regression, the aim is to estimate the coefficients by a specific formula involving matrix operations. In contrast, LASSO incorporates an additional penalty term. This term involves the sum of the absolute values of the coefficients, multiplied by a penalty parameter λ , which is a non-negative value. The inclusion of this penalty term leads to a situation where some coefficients can be reduced to zero.

LASSO regression is particularly suitable for my analysis due to its ability to simplify larger datasets which, in my case with 646 variables, is highly relevant and desired. But the primary reason is LASSO excels in preventing overfitting. In a predictive scenario, there is a real risk of the model becoming too narrowly fitted to the training data, thus ruining the forecasting capacity of the model on new datasets. The penalization approach of LASSO maintains the model's generalizability. LASSO regression is also very useful at handling a common issue in sports statistics where predictor variables are often highly correlated. LASSO addresses this by selecting one representative variable from a group of correlated ones, thereby enhancing the robustness of the model.

I have opted to use the glmnet package in R for my analysis created by Friedman, Tibshirani and Hastie (2023). For the regularization parameter in the LASSO model, I have selected

λ_{1se} . This decision is motivated by the fact that I am intending to find a model that is able to accurately predict, or at least closely approximate, the actual average rank of the game. I am using λ_{1se} , which represents the largest value of λ such that the error is within one standard error of the minimum, mainly because it is acknowledged as the industry standard while trying to maximize the predictive capability of the model, thus maintaining the norm. Furthermore, the authors of the package themselves state in the official documentation their recommendation to use λ_{1se} , even setting it as a default parameter for λ .

5 Hypotheses

This chapter introduces the initial hypotheses that will be the focus of my subsequent analysis. Unfortunately, Riot Games does not release any statistics regarding their player base, therefore every upcoming hypothesis is based on common sense, my own experience with the game and also builds on the previous chapter Effect on human performance based on time of the day. Therefore, it is reasonable these hypotheses may not reflect the results from the upcoming analysis. Their purpose is setting a reasonable framework and expectations for explaining the impact of individual hours on players' performance.

The demographic capable of significantly influencing the game's quality—either positively or negatively—must be large and consistent enough in their playtime during similar time frames. Studies covering the differences between sexes and age brackets have been made, showing significant but not large discrepancies favoring male and young players (Gagnon, 1985). An interesting fact that has been discovered by Thompson, Blair and Henrey (2014) is that around age 24 the players' response time starts to slow down but they are able to compensate through deeper game knowledge, thus maintaining their skill level.

However, none of these studies have been made on League of Legends specifically, thus reducing (but not removing) their application to my thesis. League of Legends being a MOBA, means it presents more challenges to the players than some other genres.

5.1 By demographics

I believe that the best League of Legends players could, on average, be university students. These individuals are already probably playing long enough to improve their game knowledge to a decent level, and also should be more mature than high school students, thus enabling them to keep their mental strength in check while in a match. On the other hand, they are also not old enough for their reaction speed to suffer as a consequence. They also have more free time than other people, so they can spend higher amounts of hours on the game and also presumably focus more while in a match because they have fewer worries in the back of their mind (Bath, Schilit & Lee, 2013).

I hypothesize this group of players, which could be significant—potentially even the largest—lacks a consistent gaming schedule due to the flexibility of university timetables, making it challenging to pinpoint their most active hours.

With other demographic groups, it is too hard to evaluate their proficiency only based on their lifestyle. Two other notable groups are high school students and the younger part of the workforce. I am of the opinion both of those could on average perform worse than university students, but it is too hard to determine to what extent exactly.

Formulating concrete hypotheses with regards to demographics is challenging, as there are no official public data available linking these groups and their skill level but also connecting

them to a time slot during which they prefer to play League of Legends. But even if I had that information, I believe the time slots would be too close together and too wide (most League of Legends games are played during late afternoon and evening) to use them for an analysis.

5.2 By time of the day

I believe that the most popular time slot to play League of Legends is during late afternoon, starting at 5 PM, and evening, concluding at 12 AM. I assume most games are saturated during this period, resulting in overall average quality games. As it would be surprising if the quality of matches deviated significantly during the time slot when most of them are played.

I am of the opinion that nights, more specifically between 1-6 AM, should on average possess the worst game quality. There are multiple reasons why that might be the case. Firstly, a decent part of the people playing are sleep-deprived as they usually sleep during the night. That immediately ties into another issue people playing at this time slot might encounter – mental exhaustion. I suspect a decent portion of these night players start playing during late afternoon/evening, so they are playing for a long period of time (usually without breaks), indicating their performance later in the session might drop compared to their peak level. Also, especially during Friday and Saturday, some players might be intoxicated from their evening personal life (drugs or alcohol), reducing their ability to perform during the match. On the other hand of the equation, the matchmaking process could also struggle during this time period as not many people are online, that may lead to imbalanced matches, once again reducing the game quality.

Furthermore, gaming sessions in the morning, spanning from 7 AM to 12 PM, might also fall below the average in terms of quality. This could stem from the fact that generally people have duties during the morning (school, work) so the ones that are playing could potentially be sick, thus negatively impacting their performance. The same matchmaking problem from night games could also be present here, resulting in overall quality decrease.

Next is what I conjecture to be the time slot that offers the highest quality of games – early afternoon – starting from 1 PM to 4 PM. University students, in my opinion the best demographic, should in theory be allowed to play during this timeframe thanks to their flexible schedules. I also expect a higher presence of high schoolers, but not workers as their shift usually ends at 5 PM. I believe more players should be online during this period, resulting in more balanced matchmaking.

Lastly, as I have determined every other time slot to be average or below average, one has to necessarily possess a higher game quality, and based on presented arguments I am of the opinion it is precisely early afternoon. This statement goes directly against the effect of the post lunch dip mentioned in a previous chapter. It will be interesting if its impact is large enough to invalidate other points described above.

5.3 Outline of the hypotheses

Lowest game quality during the night (1 AM – 6 AM).

Below average game quality during the morning (7 AM – 12 PM).

Above average game quality during the afternoon (1 PM – 4 PM).

Average game quality during the evening (5 PM – 12 AM).

6 Transforming ranks into integers

As stated earlier, I have chosen LASSO regression as the method for developing my model. However, LASSO necessitates the use of numeric vectors throughout the dataset. While all descriptive variables meet this criterion, the described variable – the average rank of the game – takes the form of a character variable. This forces me to modify the data to enable the use of LASSO regression.

To address this, I simply altered the names of each division into integers. Naturally, the number has to be lowest in the worst rank (Iron IV), gradually increasing with every subsequent division up to Challenger.

For the sake of simplicity, I decided to use a series of uniform integers for tiers with multiple divisions, ranging from Iron to Diamond. The sequence begins with number 1, representing the rank of Iron IV, and concludes with 28 for Diamond I, establishing a consistent progression. The reason is that these tiers function under the same set of rules; all of them allow progression by stacking league points up to 100, and players are unable to see their overall rank on the ladder.

I acknowledge that this approach is not perfect and may not reflect the reality of the skill gaps between certain divisions. However, I believe that when it comes to the balance of clarity and computational complexity, it is an appropriate and practical solution.

6.1 Apex ranks

On the other hand, I do not believe the same logic can be applied to the 3 remaining apex tiers. As these ranks represent the top of the ladder, they are unique in their way of progression.

People qualify for Master in a standard way by gaining 100 league points (LP) in Diamond I, but after that, everything changes. To promote from Master to Grandmaster, players need to be in the top 1000 highest-rated accounts on the server. This introduces a shifting threshold, usually hovering around 500 LP, that players need to surpass. Consequently, in the singular division of Master (a tier consisting of one division), players need to progress through more LP than in the whole tier of Diamond.

The same extends to the promotion from Grandmaster to Challenger, which is reserved for the top 300 players, with the cutoff roughly around 900 LP. Players also compete within Challenger, with the rank one player usually sitting near 2,000 LP, more than 1,000 LP above the threshold.

Because of the unique properties of the apex tiers, I decided to experiment with the conversion of divisions to numbers while keeping them as integers to maintain consistency and coherence. I developed 14 different variations for the rank-to-integer transformation

and established the rank prediction model, whose accuracy was then cross-validated on a different dataset.

Table 2 – The tests of errors across various model versions

Model	MSE	RMSE	RSE
1	11.0703	3.3272	0.0253
2	11.0125	3.3185	0.0252
3	11.1442	3.3383	0.0254
4	11.4959	3.3906	0.0257
5	11.9597	3.4583	0.0263
6	12.6074	3.5507	0.0270
7	13.3894	3.6592	0.0278
8	12.2668	3.5024	0.0266
9	12.1741	3.4891	0.0265
10	11.0438	3.3232	0.0252
11	11.0743	3.3278	0.0253
12	11.3940	3.3755	0.0256
13	11.4338	3.3814	0.0257
14	11.5201	3.3941	0.0258

Table 2 demonstrates that in every measured metric, model 2 exhibits the best results. The model's average prediction error, as indicated by the Root Mean Square Error (RMSE) of 3.32, is the smallest deviation from the actual rank observed in the group. Therefore, I implemented this variant in all my subsequent analyses.

The final values assigned to the apex tiers are as follows: Master at 29 (one above Diamond I), Grandmaster at 31 (two above Master), and Challenger at 33 (two above Grandmaster). It is evident that there are noticeable differences between the skill disparities within the apex tiers compared to the rest of the ladder, although they are less pronounced than I initially expected.

7 Prediction model

As shown in Model 1, the prediction model utilizes 437 variables from the dataset's 644 columns, accounting for over two-thirds of the data available. This is somewhat atypical for LASSO regression, which often significantly reduces the number of variables from a dataset. However, given that the training data consists of 41,465 observations, the model could effectively refine its parameters, despite the penalty imposed for adding extra variables.

I aimed to develop a model with the highest possible predictive accuracy. For that reason, I chose the largest value of lambda (the penalty parameter) such that the error is within one standard error of the minimum. Nonetheless, I was still under the assumption that the reduction in predictors would be more prominent.

Based on the error of this model (RMSE), I can assert that the average error in the model's predictions is about 3.32 divisions away from the actual ranks (Table 2). For making predictions, I relied solely on post-match statistics relevant to the overall game and specific roles. Additional insights could be gathered from champion picks (requiring a lot more data - around 2 million games), live match APIs, and personalized gameplay patterns of participants, including their gaming frequency, average session length, and champion choices.

Given all these factors, I am pleasantly surprised with the value of RMSE, as only a small portion of plausible data was used in a linear model, which is inherently less complex than more advanced models. There was no guarantee that just the post-match statistics would be enough to explain the average rank of the game. Therefore, I perceive the prediction model as a success.

8 Analysis of selected variables

To give readers a better understanding of the dataset and tendencies throughout the tiers, I describe some of the most significant variables. I depict the number of matches played in particular hours and rank brackets, as well as some of the meaningful predictors. The data I used for this analysis is from the whole set to give a complete and cohesive perspective.

8.1 Hour

The collection of game data across various hours was entirely random, as the API inquiry did not allow for a time specification. As Table 17 illustrates, the most popular hours to play League of Legends are between 3 PM and 12 AM, peaking at 9 PM, which marked the start of 7.65% of all matches in the dataset. This evening-centric outcome was expected, as many players view League of Legends as a leisure activity, often playing the game after completing their real-life duties, which usually occur during the morning and early afternoon.

Unsurprisingly, the lowest number of games happens during the night, with just 641, out of nearly 100,000 matches, beginning at 6 AM. This outcome seems logical, as the hour typically aligns with people's sleep schedules and the time they start their daily activities.

8.2 Rank

Table 18 displays the count of individual matches I gathered from each division. Despite specifying in my API requests to download data from players in Iron, I received a very small number of matches with said average rank. This could be attributed to Riot Games' emphasis on quickly matching players, particularly in the lower rank brackets, and Iron (especially its two lowest divisions) includes a very limited player base.

I collected a disproportionate number of matches in apex tiers as these consist of only one division. This also likely explains the high number of Diamond I games in my data, as many Master players participated in matches where the average rank of other players was lower than their own.

8.3 Variables of the basic model

To identify the most impactful variables, I developed a so-called basic model (Model 2) with an artificially limited number of predictors. This limitation was achieved by modifying the value of the penalty parameter λ in LASSO regression. Instead of setting lambda to a very minuscule value to maximize the accuracy of the prediction, I deliberately chose the value

of lambda to be 1.62. This approach allowed me to create a basic model whose main purpose is to highlight the predictors that have the greatest influence on the rank rating of a League of Legends match.

The root mean square error (RMSE) of the basic model is 5.35, which is significantly worse than the RMSE value of 3.32 of the predictive model. However, it remains sufficiently effective for conducting meaningful, albeit less precise, analysis.

In this section, I outline all the variables the basic model selected, without taking into account its coefficients. The idea is to attempt a discovery of a logical and explicable pattern across the tiers.

8.3.1 Total amount of pings per minute

Pings serve as a essential form of communication in League of Legends, allowing players to instantly convey their intentions to teammates. Unlike text chat, they can use them while controlling their champion, albeit a certain level of mechanical proficiency is necessary in this case.

Table 3 – Total amount of pings per minute per match across tiers

Tiers	Pings/minute
Challenger	19.77
Grandmaster	18.26
Master	15.38
Diamond	12.80
Emerald	10.75
Platinum	9.72
Gold	8.94
Silver	8.33
Bronze	7.85
Iron	7.14

As Table 3 indicates, the amount of pings per minute of gameplay rises with the tier. Iron players ping only 7.14 times per minute, while in a challenger game, almost 20 pings occur every minute – more than double.

This trend is primarily due to higher-tier players' proficiency in seamlessly integrating pings into gameplay. They also have a clearer idea of what they and their teammates should accomplish inside the match, thanks to their improved game knowledge.

8.3.2 Mean creep score of a top laner at 10 minutes

Creep score (CS) refers to the number of minions a top laner has collected. This metric averages the creep score of both top laners at 10 minutes of game time. This time frame corresponds to the laning phase in League of Legends, where each lane has the opportunity to farm minions, with success largely dependent on the player's skill level.

Table 4 – Mean creep score of a top laner at 10 minutes per match across tiers

Tiers	Top CS @ 10
Challenger	70.58
Grandmaster	69.59
Master	67.77
Diamond	66.22
Emerald	64.79
Platinum	63.12
Gold	61.52
Silver	59.58
Bronze	57.04
Iron	54.08

Top laners get increasingly more proficient in slaying creeps, primarily within their own lane. With every tier, there is a notable improvement, capping at over 70 CS in Challenger (Table 4). This improvement with rank is fairly logical as this statistic directly reflects the player's expertise in the game.

8.3.3 Mean amount of minions of a top laner per minute

As mentioned, players try to kill as many minions as possible to gather resources. This variable represents the average number of minions a top laner can achieve every minute. Players aim to maximize this stat to generate gold and experience.

Table 5 – Mean amount of minions of a top laner per minute per match across tiers

Tiers	Top CS/minute
Challenger	7.13
Grandmaster	7.03
Master	6.84
Diamond	6.69
Emerald	6.53
Platinum	6.32
Gold	6.12
Silver	5.88
Bronze	5.55
Iron	5.21

Table 5 reflects the top laners' ability to gather minions throughout the game. The findings closely align with those in Table 4, as they describe a similar statistic. Table 4 focuses primarily on the early game, while Table 5 takes into account the entire match. Predictably, the number of minions players secure increases the higher the average rank of the game is.

8.3.4 Mean amount of neutral minions of a jungler per minute

Junglers primarily focus on killing jungle camps, which consist of neutral minions. In doing so, they receive gold and experience.

Table 6 - Mean amount of neutral minions of a jungler per minute per match across tiers

Tiers	Jungler neutral minions/minute
Challenger	5.23
Grandmaster	5.23
Master	5.15
Diamond	5.01
Emerald	4.83
Platinum	4.68
Gold	4.55
Silver	4.39
Bronze	4.23
Iron	4.06

Table 6 clearly demonstrates that the amount of neutral minions a jungler kills increases with each tier. However, the growth per tier is not as pronounced as with other roles, as junglers need to balance farming (securing neutral minions) with ganking (assisting teammates in defeating their counterparts).

8.3.5 Mean amount of stealth wards placed by a jungler

While other roles typically purchase stealth wards, a basic trinket, at the game's start, junglers often opt for a different strategy. They often choose to start with or swap early in the game to a sweeper to deny the enemy team's vision of the map.

Table 7 - Mean amount of stealth wards placed by a jungler per match across tiers

Tiers	Jungler stealth wards
Challenger	3.04
Grandmaster	2.99
Master	2.74
Diamond	2.64
Emerald	2.96
Platinum	3.42
Gold	3.97
Silver	4.69
Bronze	5.35
Iron	5.88

Table 7 is not as straightforward to interpret as the previous tables. Junglers in lower ranks place the most stealth wards, possibly due to a lack of awareness or forgetfulness about

switching to the sweeper. The trend of reducing the quantity of stealth wards is visible up to Diamond, where it reverses and very slightly climbs back up through the apex tiers.

This may be because elite players can recognize situations in which stealth wards are valuable and choose to use them, whereas players in slightly lower tiers might switch to sweepers more routinely, without strategic consideration. The increase in placed stealth wards in the apex tiers is not drastic, as these occasions are relatively rare.

8.3.6 Mean creep score of a mid laner at 10 minutes

This statistic is a midlane parallel to the variable detailed in the section Mean creep score of a top laner at 10 minutes. Everything stated there also applies to the mid lane variant, players strive to secure as many creeps as possible.

Table 8 – Mean creep score of a mid laner at 10 minutes per match across tiers

Tiers	Mid CS @ 10
Challenger	75.01
Grandmaster	73.50
Master	71.31
Diamond	69.55
Emerald	67.64
Platinum	65.74
Gold	63.72
Silver	61.41
Bronze	58.07
Iron	54.46

Table 8 shows practically the same result as Table 4; in higher-level games, the number of minions players successfully slay increases with their rank. Interestingly, the increase in creep score at higher ranks is more notable in the mid lane than in the top lane (Table 4). There is almost a 5 CS difference between challenger mid laners and top laners.

8.3.7 Mean amount of minions of a mid laner per minute

This metric mirrors the one described in the section Mean amount of minions of a top laner per minute, but for mid laners.

Table 9 – Mean amount of minions of a mid laner per minute per match across tiers

Tiers	Mid CS/minute
Challenger	7.38
Grandmaster	7.26
Master	6.95
Diamond	6.72
Emerald	6.46
Platinum	6.21
Gold	5.98
Silver	5.74
Bronze	5.36
Iron	4.99

An interesting assertion is that both the floor and ceiling of CS per minute are more extreme in the mid lane (Table 9) than in the top lane (Table 5). I have no clear explanation as to why this may be the case. Regardless, the overall trend remains the same.

8.3.8 Mean amount of minions of a bot laner per minute

This statistic, focusing on bot laners, is the third variant of the creep score per minute analysis.

Table 10 – Mean amount of minions of a bot laner per minute per match across tiers

Tiers	Bot CS/minute
Challenger	7.63
Grandmaster	7.43
Master	7.16
Diamond	6.97
Emerald	6.68
Platinum	6.41
Gold	6.17
Silver	5.87
Bronze	5.53
Iron	5.12

Table 10 illustrates the same general trajectory as for top laners (Table 5) and mid laners (Table 9). Compared to the mid lane, creep scores per minute are approximately 0.2 creeps higher in every tier. Since the bot lane is often recognized as the role that scales up throughout the game the most, it is logical that bot laners tend to, and are allowed to, farm more than other roles.

8.3.9 Mean amount of stealth wards placed by a bot laner

Here is another statistic analogous to the previously mentioned Mean amount of stealth wards placed by a jungler, this time for bot laners. The common practice for bot laners is to swap their stealth ward for a blue orb, enhancing their safety on the map. This swap is only available after reaching level 9 in the match.

Table 11 – Mean amount of stealth wards placed by a bot laner per match across tiers

Tiers	Bot stealth wards
Challenger	4.83
Grandmaster	4.83
Master	4.81
Diamond	4.94
Emerald	5.33
Platinum	5.79
Gold	6.20
Silver	6.65
Bronze	6.94
Iron	7.11

Similarly to Table 7, Table 11 also demonstrates a general decreasing trend across tiers. Overall, the number of stealth wards placed is higher, as players need to reach level 9 before they can switch to the blue orb. There is a slight increase in the amount of stealth wards placed at Grandmaster and Challenger, suggesting that elite bot laners may sometimes prefer to use stealth wards for longer periods.

8.3.10 Mean amount of assists by a support per minute

In League of Legends, an assist occurs when a player helps teammates secure a kill but does not deliver the final blow, thereby obtaining a lesser amount of gold.

Table 12 – Mean amount of assists by a support per minute per match across tiers

Tiers	Support assists/minute
Challenger	0.48
Grandmaster	0.49
Master	0.48
Diamond	0.47
Emerald	0.45
Platinum	0.43
Gold	0.42
Silver	0.40
Bronze	0.38
Iron	0.36

There is an evident increase in assists per minute by supports as the average rank of the game increases (Table 12). I believe this phenomenon originates from two factors. Firstly, better supports try to involve themselves more on the map and look for opportunities to assist in kills. In lower tiers, supports usually just remain in the bot lane and wait for action, rather than actively seeking opportunities.

Players with a higher understanding of the game are also aware that support is usually the role least dependent on gold. Therefore, they focus on assisting with kills without taking the majority of the gold, as other roles can utilize the influx of gold more effectively.

8.3.11 Mean amount of sweeper uses by a support per minute

A sweeper is a tool used to counter enemy vision, revealing and allowing the removal of stealth (hidden) wards in the vicinity. Supports have a starting item that allows them to place stealth wards after completing a certain condition. Therefore, a sweeper is highly valuable for a support, as there is a limit on how many wards a player can place on the map.

Table 13 – Mean amount of sweeper uses by a support per minute per match across tiers

Tiers	Support sweeper/minute
Challenger	0.24
Grandmaster	0.21
Master	0.18
Diamond	0.16
Emerald	0.13
Platinum	0.11
Gold	0.10
Silver	0.08
Bronze	0.06
Iron	0.04

As Table 13 indicates, there is a significant increase in sweeper usage by supports with rank. Challenger players use the sweeper more than six times as often as an average iron support. Skilled players are able to anticipate the presence of enemy wards and also utilize their absence more effectively, which is why they use their sweeper as frequently as possible. Furthermore, some supports in the lower ranks are not even aware of the option to swap their trinket for a sweeper.

8.3.12 Mean vision score of a support per minute

Vision score is a statistic created by Riot Games to measure a player's ability to place valuable wards and destroy wards belonging to the enemy team. This metric is not tied to a single in-game action. Instead, it is based on a comprehensive scoring system created by Riot Games.

Table 14 – Mean vision score of a support per minute per match across tiers

Tiers	Support vision score/minute
Challenger	2.45
Grandmaster	2.34
Master	2.21
Diamond	2.11
Emerald	2.01
Platinum	1.90
Gold	1.78
Silver	1.64
Bronze	1.49
Iron	1.29

Table 14 shows a noticeable increase in vision score across tiers for support players. Players generally improve their warding ability as they climb through the ranks, a skill especially important for supports who contribute to the vision game the most.

8.3.13 Overview of the selected variables

After examining the variables selected by the basic model (Model 2), there is an apparent trend connecting the predictors. These factors primarily revolve around aspects of the game that players can influence regardless of other circumstances, and improving in these areas is widely regarded as beneficial for increasing the chances of winning. Creeps and wards both fulfill these conditions, as they are both advantageous and constant throughout the game without reliance on teammates.

I find it interesting, yet reasonable, that these predominantly individual statistics have the highest impact on the game. This suggests that the prowess of top-tier players may not primarily lie in the use of new and innovative tactics, but rather in their ability to perform the fundamentals of the game excellently.

9 Results

As I already have a functional predictive model, I applied it on a yet unused dataset to evaluate the quality of matches throughout the day. Like I mentioned beforehand, I intentionally selected the dataset to comprise at least 300 observations per hour, addressing an issue encountered during the early morning hours (6 AM – 8 AM). I analyzed the hours from multiple perspectives to obtain comprehensive results.

9.1 Hours with diminished and improved quality of the matches

Firstly, I conducted an Anderson-Darling normality test (Anderson & Darling, 1952) to gauge whether the difference between the prediction and actual rank exhibits a normal distribution. The result decisively indicated that the null hypothesis should be rejected, as evidenced by the very low p-value of 0.00003408. Therefore, I assume the variable "difference" does not follow a normal distribution.

After concluding the Anderson-Darling test, I decided to use the Wilcoxon test (Wilcoxon, 1945) to determine which hours enhance or, conversely, reduce the quality of the matches. The Wilcoxon test is appropriate in my case as it is a non-parametric test, meaning it does not assume any specific distribution of the data.

I then applied a one-sided Wilcoxon test on the hourly subsets, with the null hypothesis being that the median of the difference between prediction and actual rank is zero. The alternative hypothesis is either that the median of the data is greater than zero or less than zero, conducting both for each subset.

Because I performed 48 statistical tests at the 5% significance level, I adjusted the p-value with the Benjamini-Hochberg method (Benjamini & Hochberg, 1995) as a precautionary measure to reduce the likelihood of false positives. I chose this method as it is not as conservative as some other methods like the Bonferroni correction (Bonferroni, 1936), ensuring that I obtain meaningful results without being overly restrictive.

In the tables below (Table 15 and Table 16), I included only the hours which had the adjusted p-value lower than 0.05, as all my tests are at the 5% significance level. The remaining hours did not yield statistically significant results in either category.

Table 15 – Hours with a positive differences between prediction and actual rank

Hour	p-value	BH adjusted p-value
16	0.003	0.025

The only subset showing an increase in match quality encompasses matches starting at 4 PM (Table 15).

Table 16 – Hours with a negative differences between prediction and actual rank

Hour	p-value	BH adjusted p-value
1	0.001	0.020
2	0.000	0.010
3	0.000	0.010
4	0.002	0.020
5	0.004	0.031
6	0.002	0.020
7	0.006	0.037

As Table 16 illustrates, all the hours with diminished quality of the game occur consecutively during the night and early morning, from 1 AM to 7 AM. It is quite fascinating all of them happen in a sequence without even one hour being from a different time period. This shows there is certainly some sort of trend influencing the quality of the matches.

9.2 Hours compared to each other based on the quality of the matches

Similarly to the previous section, I examined whether certain hours exhibit better game quality compared to others. In this test, I again employed the Wilcoxon test, with the null hypothesis stating that the medians of the two subsets are equal. Regarding the alternative hypothesis, I only tested whether the median of the first subset is greater than the median of the second one. As I compared all hours against each other, I did not need to take into account the reverse of the alternative hypothesis, as I already covered all possibilities. I once again applied the Benjamini-Hochberg method to adjust the p-values (based on the number of tests), and selected only the pairs that showed statistical significance at the 5% level.

Table 19 shows a clear pattern, with all hours from 12 PM to 11 PM showing enhanced quality compared to the night hours of 1 AM to 7 AM. The extent to which these hours significantly outperform others depends on the individual hour. For example, hours 4 PM and 8 PM are considered to have significantly higher quality than 10 and 9 different timeslots, respectively. In contrast, 10 PM and 11 PM have a statistically significant upper hand over only one hour, in both cases 3 AM.

9.3 Means of differences between prediction and actual rank

As the last output regarding the results of my analysis, I created a plot detailing the means of differences across individual hours, providing a visual representation of the quality of League of Legends matches. I also constructed 95% confidence intervals to convey more detailed information about the gameplay quality throughout the day. The intervals were calculated using the t-test, despite my data not conforming to a normal distribution. This

decision was based on the fact that I have a substantial number of observations, ranging from hundreds to thousands, for each hour. The large sample size lends credibility to the results, as it helps mitigate the effects of non-normality.

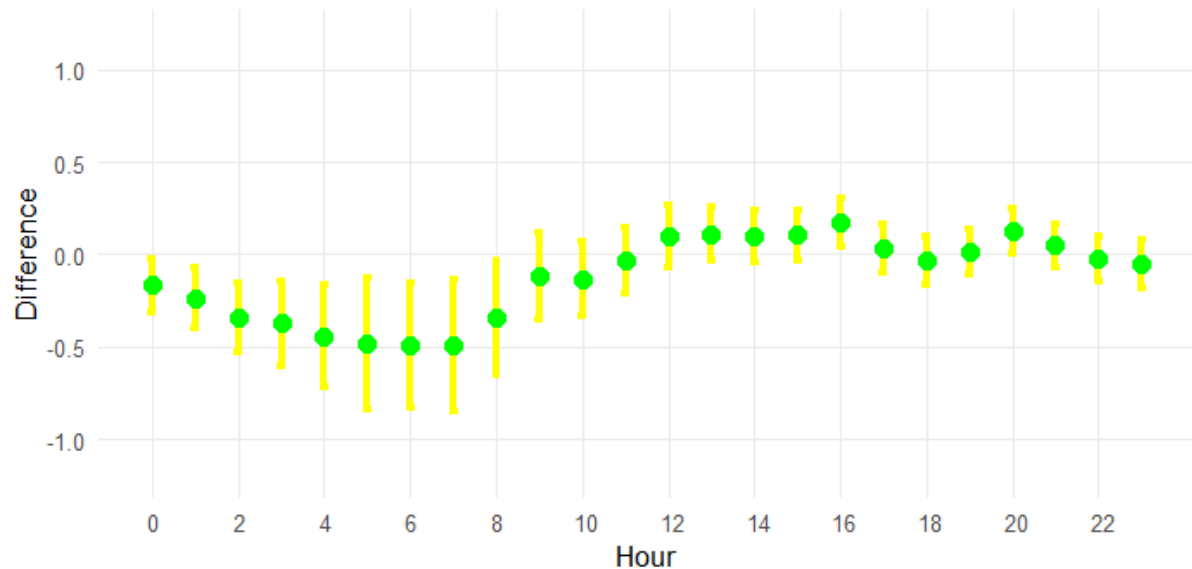


Figure 1 – Mean and its 95% confidence interval of the differences between prediction and actual rank per hour

Overall, there is a clear trend in the chart depicted in Figure 1. At 8 PM, a decline in game quality begins, lasting through the late evening and night up to 7 AM. Afterwards, the quality climbs back up, with the most substantial increase between 8 AM and 9 AM. The time period from 12 PM to 4 PM stands out as having the highest prolonged match quality during the day, peaking at 4 PM. Lastly, there is a noticeable drop during the late afternoon, but the game quality ascends once more around 8 PM, making it an isolated and second highest peak of the day.

An additional aspect to highlight regarding Figure 1 is the varying widths of the confidence intervals. Their exact values can be found in Table 20. During the nighttime, the intervals widen as I have a smaller number of observations for the nocturnal hours. On the other hand, during the afternoon and evening, I observe narrow confidence intervals because the majority of the data comes from those hours.

This ties into the observation that improvements in game quality are less pronounced compared to the declines. Matches of higher quality tend to occur during peak hours, which means that this difference is already somewhat reflected in the data. However, the small number of nighttime matches, typically of lower quality, results in a more noticeable negative difference.

10 Discussion

With the Results completed, I can now focus on rationalizing the gameplay quality of the hours and comparing them with the initial hypothesis.

As anticipated, the majority of matches occurred in the late afternoon and evening, though there was a slight variation in the start and end times of this peak period. Starting at 3 PM and ending at 11 PM, this could be influenced by the time students return from school.

The worst time slot turned out to be the nighttime, confirming the hypothesis that players at this time could be sleep-deprived, already mentally exhausted from playing a large number of games, or intoxicated. The night also turned out to be the least popular in terms of games played per hour, in line with my concerns about matchmaking quality as games are created from a wider range of ranks.

The morning period ultimately led to worsened gameplay quality compared to the base, but with notable improvement every subsequent hour. There is no clear reason why this may be the case. I suspect a higher percentage of medically indisposed people play during this time, as they might be more likely to stay home instead of attending their daily activities. The quality growth between the individual hours during this time slot might be triggered as people wake up. Since people are often groggy after waking up, waiting a few hours before playing might lead to better performance in the game.

Afterwards, the most enhanced match quality time interval occurs – early afternoon. I correctly anticipated this time period to possess high quality matches, based on the fact that a lot of students are already back from school, allowing them to play without neglecting their responsibilities. Additionally, the majority of workers have not finished their jobs yet, a demographic that I believe contributes to the decline in match quality.

The last and most popular time slot is the late afternoon/evening. Even this span of hours generally adheres to my hypothesis of average game quality, with a few exceptions. I expected the differences between prediction and actual rank to be more constant and start to decline around 10 PM as players get more tired and exhausted. This did not turn out to be the case, as there is a substantial drop at 6 PM, immediately followed by a subsequent rise, peaking at 8 PM, the hour with the second-highest enhanced quality of the whole day. I can rationalize the 6 PM decline as people from the workforce return home and start playing, dragging the quality of the matches down. However, I have no reasonable explanation for the observed rise in quality at 8 PM. The rest of the evening is quite straightforward, with the difference between prediction and actual rank descending every hour until midnight, as players' fatigue accumulated throughout the day settles in.

In general, my hypothesis proved to be quite accurate. After concluding the analysis, I can state there is an evident effect of the time of day on match quality in League of Legends, indicating that certain hours significantly influence the overall gameplay experience.

10.1 Practical applications

I believe that League of Legends players can derive practical usage from my study in two main ways. Firstly, if players seek highly competitive matches where they can be challenged by their opponents while maintaining their rank, they should consider playing at 4 PM or 8 PM. In these time slots, they will, on average, experience elevated game quality. That can enable them to improve at a faster pace as playing against more skilled players is beneficial for their development as a League of Legends player.

On the other hand, if players aspire to achieve as high of a rank as possible, playing during the nighttime, especially from 3 AM to 6 AM, could be favourable. As these hours feature some of the worst game quality throughout the day, players are matched against competitors who are on the same level rank-wise, but their actual skill during these night matches is lower. This advantage could allow them to climb with more ease and attain their dream rank goal.

Conclusions

The thesis focused on how different hours throughout the day impact the quality of matches in the popular multiplayer online game, League of Legends. Through the analysis of a comprehensive dataset and the application of advanced statistical methods, I uncovered insightful findings. These not only contribute to the understanding of online gaming dynamics but also offer practical implications for players and game developers.

In this study, data were successfully gathered using Riot Games' API, providing access to a large number of post-game metrics on individual League of Legends matches. However, this process faced challenges, especially due to the API's limitations and technical issues. The dataset formed the backbone of the analysis, allowing for a detailed examination of various aspects influencing match quality. Therefore, my thesis can serve as guidance for generating recent data or as an exemplary dataset with up-to-date matches.

The technical aspect of the study, involving LASSO with the help of transformation of ranks into integers, proved effective in modeling and predicting match quality. This method enabled the identification of a predictive model, intended to maximize the accuracy of the predictions, and also a basic model, whose purpose is to select the most important variables throughout the dataset. The aforementioned approach can serve as a prototype for similar studies in other multiplayer online games.

Interestingly, the most significant variables, as selected by the basic model, show a great emphasis on variables a certain player can directly influence without reliance on their teammates. They mainly revolve around creep score and vision, categories which are generally very beneficial to progressing the game in an autonomous way. This focus on core gameplay elements highlights that the main differences in players' skill lie in their ability to execute the fundamental parts of the game.

My research findings demonstrate a significant influence of the time of day on match quality. This effect is rooted in the varying mental and physical states of players during the 24-hour period, mainly caused by the circadian rhythm. The results show that the early afternoon, particularly around 4 PM, offers the highest quality of gameplay. This may be due to the players' peak alertness and the increased presence of presumably above-average skilled university students in the player base. On the contrary, the late-night and early morning hours, specifically between 1 AM and 7 AM, are characterized by a notable dip in match quality. This can be linked to factors such as fatigue, sleep deprivation, and possibly the effects of nocturnal lifestyle choices.

The research led to a variety of practical implications. For players looking to enhance their skills or enjoy more competitive matches, timing their sessions according to the identified peak hours could be beneficial. Conversely, those aiming to climb the game's ranking ladder might find it advantageous to play during off-peak hours, where the match quality is relatively lower. For game developers, understanding these time-related dynamics could

assist in decisions regarding matchmaking algorithms, possibly altering them throughout the day.

Overall, my thesis provided insight on the relationship between the time of day and match quality in League of Legends. It underscores the importance of biological rhythms and lifestyle patterns with respect to players' performance in online gaming environments. While the study is focused on League of Legends, its methodology and findings have broader implications. I suspect that similar patterns could be observed in other online multiplayer games. Future research could expand on this work by exploring some other factors influencing match quality, such as seasonal variations or even the impact of global events.

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Appendices

Appendix A: Models

Model 1 – Prediction model

prediction =	
-25.8207	(Intercept)
-3.80022	server_KR
1.15878	server_NA
-1.51692	server_VN
0.06366	blastcone_enemy_total
-0.08307	clean_aces_total
-0.0004	dodged_skills
0.00744	drake_delta
0.20793	drake_total
-12.1489	drake_total_per_min
0.30219	duration
0.0043	early_adv_delta
0.09296	early_adv_total
0.06876	elder_drake_delta
-0.16761	elder_drake_total
-0.04755	epics_flips_total
0.01819	epics_on_spawn_total
-0.01666	epics_steals_smiteless_total
0.01917	first_drake_time

-0.0601	first_drake_win
0.05013	first_herald_win
-0.03715	first_inhibitor_win
-0.07354	first_turret_time
0.20496	herald_multi_turret_total
-0.08019	herald_total
0.07552	champ_select_changes
-0.06857	inhibitor_delta
1.01697	inhibitor_total_per_min
-0.9484	invade_kills_TF
-0.02273	nash_delta
0.93345	nash_total
0.10643	perfect_soul
0.31781	pings_per_min
-0.00316	pings_total
-0.02163	players_on_expected_position_total
-0.30268	surrender
0.0642	survived_on_1HP_total
0.06594	takedowns_in_enemy_fountain
0.01114	toxic_pings_total
0.00584	turret_delta
-0.10948	turret_pre_14_TF
-0.02097	turret_total
-0.11449	turrets_taken_with_herald
0.01347	vision_pings_total

-0.00183	top_ally_jg_minions_delta
0.01143	top_assists_avg
1.40997	top_assists_avg_per_min
-2.63654	top_control_wards_placed_avg_per_min
0.02276	top_deaths_avg
0.86476	top_deaths_avg_per_min
-0.00321	top_deaths_delta
-0.00001	top_dmg_taken_avg
-0.00143	top_dmg_taken_avg_per_min
0	top_dmg_taken_delta
-0.00018	top_dmg_to_buildings_avg
0.00002	top_dmg_to_buildings_delta
0.00296	top_dmg_to_champions_avg_per_min
0.00004	top_dmg_to_objectives_avg
0	top_dmg_to_objectives_delta
-0.10745	top_double_kills_avg
0.16676	top_enemy_jg_minions_avg_per_min
0.00402	top_enemy_jg_minions_delta
0.1146	top_first_blood_assist
-0.07772	top_first_blood_kill
-0.02923	top_first_turret_assist_for_winning_team
-0.14785	top_first_turret_kill_for_winning_team
-0.05175	top_first_turret_takedown_tf
0.00388	top_gold_earned_avg_per_min
0	top_gold_earned_delta

1.76618	top_kill_part_avg
-0.10904	top_kills_after_hidden_with_ally_avg_per_min
0.0027	top_kills_avg
-0.00292	top_kills_delta
0.07758	top_kills_near_enemy_turret_avg
0.44622	top_kills_on_other_lanes_avg
0.11612	top_kills_under_own_turret_avg
0.04317	top_lane_minions_10min_avg
0.0147	top_largest_killing_spree_avg
-0.08835	top_level_avg
-12.9958	top_level_avg_per_min
-0.00041	top_longest_time_alive_avg
-0.01385	top_minions_avg
1.35363	top_minions_avg_per_min
-0.00052	top_minions_delta
1.00746	top_neutral_minions_avg_per_min
0.02551	top_plates_takedowns_avg
0.0046	top_plates_takedowns_delta
-0.72578	top_quadra_kills_avg
-0.18779	top_quick_first_turret_TF
0.0393	top_scuttle_kills_avg
0.00195	top_scuttle_kills_delta
-0.67876	top_solo_kills_avg_per_min
-0.01269	top_solo_kills_delta
-2.44146	top_solo_nash_avg

-0.1209	top_stealth_wards_placed_avg
-0.0035	top_stealth_wards_placed_delta
-0.08061	top_triple_kills_avg
0.26042	top_turrets_avg
-6.70257	top_turrets_avg_per_min
0.00524	top_turrets_delta
0.00308	top_vision_score_avg
3.1039	top_vision_score_avg_per_min
1.3305	top_wards_guarded_avg_per_min
0.05369	top_wards_guarded_delta
-0.61859	top_wards_takedowns_avg_per_min
-0.00242	top_wards_takedowns_delta
-0.39724	tops_percentage_of_team_ally_jg_minions_avg
0.08251	tops_percentage_of_team_assists_avg
0.13826	tops_percentage_of_team_deaths_avg
0.75868	tops_percentage_of_team_dmgtaken_avg
0.2504	tops_percentage_of_team_dmgtobuildings_avg
-1.73305	tops_percentage_of_team_dmgtobjectives_avg
0.08353	tops_percentage_of_team_solo_kills_avg
2.6423	tops_percentage_of_team_stealth_wards_placed_avg
0.44438	tops_percentage_of_team_turrets_avg
-4.76214	tops_percentage_of_team_vision_score_avg
0.00179	jungle_ally_jg_minions_avg
-0.00645	jungle_ally_jg_minions_delta
2.02797	jungle_assists_avg_per_min

-0.00012 jungle_assists_delta

-0.1967 jungle_control_wards_placed_avg

-0.00685 jungle_control_wards_placed_delta

0.00514 jungle_deaths_avg

0.71614 jungle_deaths_avg_per_min

0 jungle_dmg_taken_delta

-0.00654 jungle_dmg_to_buildings_avg_per_min

0.00003 jungle_dmg_to_buildings_delta

0.00338 jungle_dmg_to_champions_avg_per_min

0 jungle_dmg_to_champions_delta

0.00001 jungle_dmg_to_objectives_avg

0 jungle_dmg_to_objectives_delta

-0.19664 jungle_double_kills_avg

-0.05809 jungle_enemy_jg_minions_avg

2.55762 jungle_enemy_jg_minions_avg_per_min

0.00233 jungle_enemy_jg_minions_delta

0.26817 jungle_first_blood_assist

0.02183 jungle_first_blood_kill

0.05404 jungle_first_blood_kill_for_winning_team

-0.10033 jungle_first_turret_assist_for_winning_team

-0.11547 jungle_first_turret_kill_for_winning_team

-0.10618 jungle_first_turret_takedown_tf

0.0035 jungle_gold_earned_avg_per_min

-0.00001 jungle_gold_earned_delta

-0.01639 jungle_initial_crab_delta

-0.06647	jungle_jg_kills_near_dmged_epic_avg
-0.01172	jungle_jungle_cs_10min_avg
0.00181	jungle_jungle_cs_10min_delta
-0.00091	jungle_kda_avg
0.00732	jungle_kda_delta
0.20546	jungle_kill_part_avg
0.00541	jungle_kills_after_hidden_with_ally_avg
1.72957	jungle_kills_after_hidden_with_ally_avg_per_min
0.00507	jungle_kills_delta
0.18295	jungle_kills_early_jungle_avg
-0.08481	jungle_kills_early_jungle_delta
0.12121	jungle_kills_laners_early_avg
0.11336	jungle_kills_near_enemy_turret_avg
0.13472	jungle_kills_under_own_turret_avg
0.12279	jungle_lane_minions_10min_avg
-0.00921	jungle_lane_minions_10min_delta
0.04223	jungle_largest_killing_spree_avg
-6.54625	jungle_level_avg_per_min
-0.0084	jungle_level_delta
-0.0003	jungle_longest_time_alive_avg
-0.03669	jungle_minions_avg
1.17973	jungle_minions_avg_per_min
-0.0007	jungle_minions_delta
1.67008	jungle_neutral_minions_avg_per_min
0.01459	jungle_objectives_stolen_avg

-0.33316	jungle_penta_kills_avg
0.15946	jungle_plates_takedowns_avg
-0.13029	jungle_quadra_kills_avg
-0.13672	jungle_quick_first_turret_TF
-0.04666	jungle_scuttle_kills_avg
9.21126	jungle_scuttle_kills_avg_per_min
-0.5773	jungle_solo_kills_avg_per_min
-0.00672	jungle_solo_kills_delta
-1.03783	jungle_solo_nash_avg
-4.13218	jungle_stealth_wards_placed_avg_per_min
0.00454	jungle_stealth_wards_placed_delta
10.37409	jungle_sweeper_uses_avg_per_min
-0.08021	jungle_triple_kills_avg
0.23307	jungle_turrets_avg
-0.0177	jungle_turrets_delta
0.0273	jungle_vision_score_avg
0.25134	jungle_vision_score_avg_per_min
0.00222	jungle_vision_score_delta
0.00595	jungle_wards_guarded_avg
1.69864	jungle_wards_guarded_avg_per_min
0.01339	jungle_wards_guarded_delta
0.05023	jungle_wards_takedowns_avg
-0.00074	jungle_wards_takedowns_delta
0.48895	jungles_percentage_of_team_assists_avg
0.53958	jungles_percentage_of_team_control_wards_placed_avg

-6.96812	jungles_percentage_of_team_dmg_taken_avg
0.30123	jungles_percentage_of_team_dmg_to_buildings_avg
1.49152	jungles_percentage_of_team_dmg_to_objectives_avg
-0.04205	jungles_percentage_of_team_solo_kills_avg
-1.99472	jungles_percentage_of_team_stealth_wards_placed_avg
0.01812	jungles_percentage_of_team_turrets_avg
-0.12357	jungles_percentage_of_team_wards_takedowns_avg
-0.02057	mid_ally_jg_minions_avg
0.21263	mid_ally_jg_minions_avg_per_min
-0.00191	mid_ally_jg_minions_delta
0.00046	mid_assists_avg
2.59288	mid_assists_avg_per_min
-0.00168	mid_assists_delta
0.71037	mid_control_wards_placed_avg_per_min
0.01032	mid_control_wards_placed_delta
-0.03136	mid_deaths_avg
0.00592	mid_deaths_delta
-0.0005	mid_dmg_taken_avg
0	mid_dmg_taken_delta
-0.00329	mid_dmg_to_buildings_avg_per_min
0.00001	mid_dmg_to_buildings_delta
0.00356	mid_dmg_to_champions_avg_per_min
0	mid_dmg_to_champions_delta
0.00001	mid_dmg_to_objectives_avg
-0.16838	mid_double_kills_avg

0.0197	mid_enemy_jg_minions_avg_per_min
0.00032	mid_enemy_jg_minions_delta
0.29916	mid_first_blood_assist
0.02413	mid_first_blood_kill
-0.09709	mid_first_turret_assist
0.07275	mid_first_turret_assist_for_winning_team
-0.08229	mid_first_turret_kill_for_winning_team
-0.06747	mid_first_turret_takedown_tf
0.00696	mid_gold_earned_avg_per_min
-0.00001	mid_gold_earned_delta
-0.00865	mid_kda_avg
0.20724	mid_kill_part_avg
-0.44085	mid_kills_after_hidden_with_ally_avg_per_min
0.00745	mid_kills_delta
0.0648	mid_kills_near_enemy_turret_avg
0.40259	mid_kills_on_other_lanes_avg
0.0831	mid_kills_under_own_turret_avg
0.03946	mid_lane_minions_10min_avg
0.00168	mid_lane_minions_10min_delta
0.0102	mid_largest_killing_spree_avg
-0.72967	mid_level_avg
-0.00023	mid_longest_time_alive_avg
-0.0389	mid_minions_avg
2.23668	mid_minions_avg_per_min
-0.00142	mid_minions_delta

-0.00401	mid_neutral_minions_avg
0.43268	mid_neutral_minions_avg_per_min
0.00339	mid_plates_takedowns_avg
-0.00224	mid_plates_takedowns_delta
-0.00895	mid_quadra_kills_avg
0.84498	mid_quick_first_turret_TF
0.01821	mid_scuttle_kills_delta
0.00151	mid_solo_kills_avg
-0.60075	mid_solo_kills_avg_per_min
-0.00643	mid_solo_kills_delta
-0.50842	mid_solo_nash_avg
-0.08081	mid_stealth_wards_placed_avg
-3.04578	mid_stealth_wards_placed_avg_per_min
-0.01029	mid_stealth_wards_placed_delta
0.24805	mid_takedowns_after_lvl_adv_avg
-0.39033	mid_triple_kills_avg
-1.76959	mid_turrets_avg_per_min
0.03879	mid_turrets_delta
1.82668	mid_vision_score_avg_per_min
-0.00176	mid_vision_score_delta
0.87582	mid_wards_guarded_avg_per_min
-0.25384	mid_wards_takedowns_avg_per_min
0.00074	mid_wards_takedowns_delta
-0.00056	mids_percentage_of_team_control_wards_placed_avg
0.08749	mids_percentage_of_team_deaths_avg

4.74498	mids_percentage_of_team_dmg_taken_avg
-0.98097	mids_percentage_of_team_dmg_to_buildings_avg
-0.09325	mids_percentage_of_team_dmg_to_champions_avg
-0.19371	mids_percentage_of_team_dmg_to_objectives_avg
-0.05394	mids_percentage_of_team_drake_kills_avg
1.00641	mids_percentage_of_team_kills_avg
0.01418	mids_percentage_of_team_minions_avg
3.45814	mids_percentage_of_team_neutral_minions_avg
-1.43837	mids_percentage_of_team_stealth_wards_placed_avg
0.52116	mids_percentage_of_team_turrets_avg
1.04961	mids_percentage_of_team_vision_score_avg
-0.06478	mids_percentage_of_team_wards_takedowns_avg
1.07999	bot_ally_jg_minions_avg_per_min
-0.00151	bot_ally_jg_minions_delta
0.02622	bot_assists_avg
0.00049	bot_assists_delta
-0.02109	bot_control_wards_placed_avg
0.00597	bot_control_wards_placed_delta
1.93167	bot_deaths_avg_per_min
-0.00001	bot_dmg_taken_avg
0.00001	bot_dmg_taken_delta
-0.00013	bot_dmg_to_buildings_avg
-0.00072	bot_dmg_to_buildings_avg_per_min
0.00002	bot_dmg_to_buildings_delta
0.00427	bot_dmg_to_champions_avg_per_min

0	bot_dmg_to_champions_delta
0.00004	bot_dmg_to_objectives_avg
0.00046	bot_dmg_to_objectives_avg_per_min
0	bot_dmg_to_objectives_delta
-0.16962	bot_double_kills_avg
1.68162	bot_enemy_jg_minions_avg_per_min
-0.00317	bot_enemy_jg_minions_delta
0.0341	bot_first_blood_assist
0.04464	bot_first_turret_assist
0.05311	bot_first_turret_takedown_tf
0.00261	bot_gold_earned_avg_per_min
0	bot_gold_earned_delta
0.04818	bot_jg_kills_near_dmged_epic_avg
0.03362	bot_kills_after_hidden_with_ally_avg
-0.03499	bot_kills_avg
-1.31677	bot_kills_avg_per_min
0	bot_kills_delta
0.19419	bot_kills_on_other_lanes_avg
0.1026	bot_kills_under_own_turret_avg
0.0644	bot_lane_minions_10min_avg
-0.0011	bot_lane_minions_10min_delta
0.0163	bot_largest_killing_spree_avg
-0.32513	bot_level_avg
0.04281	bot_level_avg_per_min
0.01805	bot_level_delta

-0.0005	bot_longest_time_alive_avg
-0.0276	bot_minions_avg
2.51226	bot_minions_avg_per_min
-0.01405	bot_neutral_minions_avg
0.20136	bot_plates_takedowns_avg
0.01611	bot_plates_takedowns_delta
0.10046	bot_quadra_kills_avg
0.34199	bot_quick_first_turret_TF
0.15818	bot_scuttle_kills_avg
0.46553	bot_scuttle_kills_avg_per_min
0.62104	bot_solo_kills_avg_per_min
-0.05024	bot_solo_kills_delta
-0.6003	bot_solo_nash_avg
-0.41866	bot_stealth_wards_placed_avg
0.00105	bot_stealth_wards_placed_delta
0.15626	bot_takedowns_after_lvl_adv_avg
0.02553	bot_turrets_delta
0.01905	bot_vision_score_avg
0.51115	bot_vision_score_avg_per_min
0.01183	bot_wards_guarded_delta
-0.04568	bot_wards_takedowns_avg
-0.13973	bot_wards_takedowns_avg_per_min
-0.00001	bot_wards_takedowns_delta
1.66668	bots_percentage_of_team_ally_jg_minions_avg
-0.60653	bots_percentage_of_team_control_wards_placed_avg

-1.05724	bots_percentage_of_team_dmg_to_buildings_avg
0.42277	bots_percentage_of_team_dmg_to_objectives_avg
0.2169	bots_percentage_of_team_drake_kills_avg
-10.2446	bots_percentage_of_team_minions_avg
0.42636	bots_percentage_of_team_plates_takedowns_avg
5.9526	bots_percentage_of_team_stealth_wards_placed_avg
0.52328	bots_percentage_of_team_turrets_avg
7.07021	bots_percentage_of_team_vision_score_avg
0.23164	bots_percentage_of_team_wards_takedowns_avg
-0.15443	supp_ally_jg_minions_avg_per_min
0.00186	supp_ally_jg_minions_delta
0.34025	supp_assists_avg_per_min
0.0049	supp_assists_delta
-0.14099	supp_control_wards_placed_avg
0.00345	supp_deaths_delta
-0.00001	supp_dmg_taken_avg
0	supp_dmg_taken_avg_per_min
0.00001	supp_dmg_taken_delta
-0.00001	supp_dmg_to_buildings_avg
-0.00167	supp_dmg_to_buildings_avg_per_min
0.00002	supp_dmg_to_buildings_delta
0.00455	supp_dmg_to_champions_avg_per_min
0.00001	supp_dmg_to_champions_delta
0.00005	supp_dmg_to_objectives_avg
-0.00001	supp_dmg_to_objectives_delta

0.78801	supp_enemy_jg_minions_avg_per_min
0.00087	supp_enemy_jg_minions_delta
0.12101	supp_fast_quest_completion_avg
-0.06801	supp_faster_quest_completion_for_winning_team
0.03971	supp_first_blood_assist
0.00176	supp_first_blood_assist_for_winning_team
-0.07859	supp_first_turret_assist_for_winning_team
-0.12411	supp_first_turret_kill
-0.03907	supp_first_turret_takedown_tf
0.01884	supp_gold_earned_avg_per_min
-0.07807	supp_jg_kills_near_dmged_epic_avg
0.00043	supp_kda_avg
1.92153	supp_kill_part_avg
0.01508	supp_kills_after_hidden_with_ally_avg
0.53059	supp_kills_after_hidden_with_ally_avg_per_min
-0.01008	supp_kills_avg
-6.55702	supp_kills_avg_per_min
-0.00198	supp_kills_delta
0.1042	supp_kills_near_enemy_turret_avg
0.44183	supp_kills_on_other_lanes_avg
0.08793	supp_kills_under_own_turret_avg
0.03517	supp_lane_minions_10min_avg
0.03166	supp_largest_killing_spree_avg
-0.05498	supp_level_avg
-10.4759	supp_level_avg_per_min

-0.06605	supp_level_delta
-0.00001	supp_longest_time_alive_avg
-0.00881	supp_minions_avg
-0.00035	supp_minions_delta
-2.33955	supp_penta_kills_avg
-0.07449	supp_plates_takedowns_avg
-0.00369	supp_plates_takedowns_delta
0.08092	supp_quadra_kills_avg
0.32142	supp_quick_first_turret_TF
-0.02519	supp_scuttle_kills_avg
-4.82815	supp_scuttle_kills_avg_per_min
-0.00509	supp_scuttle_kills_delta
0.13024	supp_solo_kills_avg
4.07193	supp_solo_kills_avg_per_min
-0.01835	supp_solo_kills_delta
7.93587	supp_solo_nash_avg
-0.34779	supp_stealth_wards_placed_avg_per_min
5.8314	supp_sweeper_uses_avg_per_min
0.23257	supp_takedowns_after_lvl_adv_avg
0.05233	supp_triple_kills_avg
0.10973	supp_turrets_avg
0.03682	supp_vision_score_avg
1.43288	supp_vision_score_avg_per_min
0.00067	supp_vision_score_delta
-0.004	supp_wards_guarded_avg

-0.01929	supp_wards_takedowns_avg
-0.63647	supp_wards_takedowns_avg_per_min
-0.2696	supps_own_percentage_of_kills_after_hidden_with_ally_avg
-0.89349	supps_percentage_of_team_ally_jg_minions_avg
-0.42455	supps_percentage_of_team_assists_avg
0.47708	supps_percentage_of_team_control_wards_placed_avg
-1.56055	supps_percentage_of_team_deaths_avg
-9.893	supps_percentage_of_team_dmg_taken_avg
-4.56072	supps_percentage_of_team_dmg_to_champions_avg
-1.03896	supps_percentage_of_team_dmg_to_objectives_avg
-0.3805	supps_percentage_of_team_drake_kills_avg
-1.33541	supps_percentage_of_team_gold_earned_avg
-0.72025	supps_percentage_of_team_kills_avg
5.56633	supps_percentage_of_team_neutral_minions_avg
-0.23463	supps_percentage_of_team_plates_takedowns_avg
-0.06459	supps_percentage_of_team_solo_kills_avg
-0.15275	supps_percentage_of_team_turrets_avg
-0.10816	supps_percentage_of_team_wards_takedowns_avg

Model 2 – Basic model

prediction =		
-13.8725	(Intercept)	
0.437502	pings_per_min	
0.032993	top_lane_minions_10min_avg	
0.456693	top_minions_avg_per_min	
1.057581	jungle_neutral_minions_avg_per_min	
-0.01225	jungle_stealth_wards_placed_avg	
0.018853	mid_lane_minions_10min_avg	
0.919724	mid_minions_avg_per_min	
1.442112	bot_minions_avg_per_min	
-0.00053	bot_stealth_wards_placed_avg	
0.130027	supp_assists_avg_per_min	
2.339508	supp_sweeper_uses_avg_per_min	
0.880252	supp_vision_score_avg_per_min	

Appendix B: Tables

Table 17 – Number of hours per hour

Hour	# of matches	% of matches
0	4994	5.06%
1	3848	3.90%
2	2858	2.89%
3	1970	2.00%
4	1309	1.33%

5	827	0.84%
6	641	0.65%
7	762	0.77%
8	1178	1.19%
9	1819	1.84%
10	2546	2.58%
11	3130	3.17%
12	3797	3.85%
13	4552	4.61%
14	5295	5.36%
15	6004	6.08%
16	6136	6.22%
17	6181	6.26%
18	6045	6.12%
19	6548	6.63%
20	7342	7.44%
21	7550	7.65%
22	7171	7.26%
23	6223	6.30%

Table 18 – Number of games per division

Tier	# of matches	% of matches
Challenger	2294	2.32%
Grandmaster	6105	6.18%
Master	11450	11.60%
Diamand I	6161	6.24%
Diamand II	3618	3.66%
Diamand III	3124	3.16%
Diamand IV	2392	2.42%
Emerald I	2266	2.30%
Emerald II	2721	2.76%
Emerald III	4006	4.06%
Emerald IV	4657	4.72%
Platinum I	3544	3.59%
Platinum II	3174	3.21%
Platinum III	3225	3.27%
Platinum IV	3202	3.24%
Gold I	2921	2.96%
Gold II	2505	2.54%
Gold III	2671	2.71%
Gold IV	2375	2.41%
Silver I	2130	2.16%
Silver II	2465	2.50%
Silver III	3719	3.77%
Silver IV	3771	3.82%

Bronze I	2815	2.85%
Bronze II	2186	2.21%
Bronze III	2211	2.24%
Bronze IV	2803	2.84%
Iron I	3346	3.39%
Iron II	831	0.84%
Iron III	38	0.04%
Iron IV	0	0.00%

Table 19 – Comparison of hours against each other based on the quality of the matches

Hour 1	Greater then	Hour 2	p-value	BH adjusted p-value
12	>	1	0.002	0.029
12	>	2	0.001	0.016
12	>	3	0.001	0.016
12	>	4	0.002	0.024
12	>	5	0.003	0.031
12	>	6	0.002	0.024
12	>	7	0.004	0.039
13	>	0	0.006	0.050
13	>	1	0.001	0.016
13	>	2	0.000	0.009
13	>	3	0.000	0.009
13	>	4	0.001	0.016
13	>	5	0.002	0.024
13	>	6	0.001	0.018
13	>	7	0.003	0.030
14	>	0	0.005	0.048
14	>	1	0.001	0.016
14	>	2	0.000	0.009
14	>	3	0.000	0.009
14	>	4	0.001	0.016
14	>	5	0.002	0.024
14	>	6	0.001	0.018
14	>	7	0.003	0.030
15	>	0	0.006	0.048
15	>	1	0.001	0.016
15	>	2	0.000	0.009
15	>	3	0.000	0.009
15	>	4	0.001	0.016
15	>	5	0.002	0.024
15	>	6	0.001	0.018

15	>	7	0.003	0.031
16	>	0	0.000	0.009
16	>	1	0.000	0.005
16	>	2	0.000	0.002
16	>	3	0.000	0.002
16	>	4	0.000	0.007
16	>	5	0.000	0.011
16	>	6	0.000	0.009
16	>	7	0.001	0.016
16	>	8	0.001	0.021
16	>	10	0.003	0.031
17	>	2	0.003	0.035
17	>	3	0.002	0.027
17	>	4	0.006	0.048
17	>	6	0.006	0.048
18	>	3	0.005	0.046
19	>	2	0.002	0.029
19	>	3	0.001	0.021
19	>	4	0.004	0.039
19	>	6	0.004	0.039
20	>	0	0.002	0.024
20	>	1	0.000	0.009
20	>	2	0.000	0.007
20	>	3	0.000	0.007
20	>	4	0.000	0.013
20	>	5	0.001	0.018
20	>	6	0.001	0.016
20	>	7	0.002	0.025
20	>	8	0.004	0.039
21	>	1	0.003	0.035
21	>	2	0.001	0.018
21	>	3	0.001	0.016
21	>	4	0.002	0.029
21	>	5	0.004	0.039
21	>	6	0.003	0.031
22	>	3	0.004	0.037
23	>	3	0.005	0.043

Table 20 – Mean and its 95% confidence interval of the differences between prediction and actual rank per hour

Hour	Mean	Lower CI	Upper CI
0	-0.17	-0.31	-0.02
1	-0.24	-0.40	-0.07
2	-0.34	-0.53	-0.15
3	-0.37	-0.61	-0.14
4	-0.44	-0.72	-0.16
5	-0.48	-0.84	-0.13
6	-0.49	-0.83	-0.16
7	-0.49	-0.85	-0.14
8	-0.34	-0.65	-0.04
9	-0.12	-0.36	0.12
10	-0.13	-0.34	0.07
11	-0.03	-0.21	0.15
12	0.10	-0.07	0.26
13	0.11	-0.04	0.26
14	0.10	-0.04	0.24
15	0.10	-0.03	0.24
16	0.17	0.05	0.30
17	0.03	-0.10	0.16
18	-0.03	-0.16	0.10
19	0.01	-0.11	0.14
20	0.13	0.01	0.25
21	0.05	-0.07	0.17
22	-0.03	-0.15	0.09
23	-0.05	-0.18	0.08

Appendix C: Dataset

Included as a supplementary file "Appendix C.zip" with the thesis.

Appendix D: R Scripts

Included as a supplementary file "Appendix D.zip" with the thesis.